

AI for Volumetric Exploration in Reality (**AIVER**) – Real-Time Virtual Reality Visualization of Unstructured Volumes Using Gaussian Splatting and AI-Accelerated Rendering

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Project Summary

Immersive visualization is becoming essential for exploring large unstructured volumetric datasets across computational science, engineering, and experimental data-centric research. Yet, existing GPU-accelerated volume-rendering systems struggle to maintain interactive performance, especially in immersive environments such as Virtual Reality (VR), where modern head-mounted displays require low latency and high stereo stability. Gaussian splatting has recently emerged as a powerful technique for real-time rendering of neural radiance fields and point-based scene representations. However, no existing framework applies Gaussian splats to unstructured scientific grids, nor integrates them into a real-time VR pipeline that supports transfer-function editing, spatial manipulation, or multivariate visual analysis. As a result, scientists lack an immersive, low-latency environment for exploring complex datasets at full resolution.

This project addresses five critical gaps in prior work: **(1)** the absence of a VR framework for unstructured Gaussian volumes; **(2)** the lack of multivariate, transfer-function-agnostic Gaussian representations; **(3)** the difficulty of sustaining stereo frame-rate stability in splat-based methods; **(4)** the absence of illumination and gradient-aware shading models for Gaussian-encoded scientific fields; and **(5)** the lack of AI-accelerated pipelines for Gaussian volume rendering. Together, these gaps prevent immersive, real-time analysis of unstructured simulation data and limit the scalability of next-generation scientific workflows.

To overcome these limitations, the project pursues four tightly aligned goals. **Goal 1** is to develop a real-time VR rendering system for unstructured volumes using Gaussian splatting integrated with the Open Volume Renderer (OVR) and OpenXR. This system will provide low-latency, stereo-stable visualization of large unstructured datasets. **Goal 2** is to create a volume-to-Gaussian transcoder that converts tetrahedral meshes, irregular grids, and multiresolution structures into optimized Gaussian primitives suitable for VR. **Goal 3** introduces interactive spatial manipulation (e.g. rotation, slicing, scaling, and region-of-interest selection) directly on the Gaussian representation for enhanced scientific exploration. **Goal 4** establishes real-time seamless visualization with support for multivariate data, allowing users to manipulate color and opacity mappings on the fly without reprocessing or retraining.

The expected outcomes include: a functional VR engine for unstructured Gaussian volumes; a deployable unstructured-to-Gaussian transcoder; a complete VR interaction module for manipulating Gaussian datasets; a multivariate transfer-function system; a low-latency stereo Gaussian renderer; prototype gradient-aware shading models; and a foundational AI-enabled acceleration pipeline for Gaussian pruning, clustering, and neural sampling. These deliverables will be integrated into a unified, well-documented software package.

This proposed project will establish the scientific and technical foundation for a multi-year research agenda on AI-accelerated Gaussian representations for unstructured data. By combining VR immersion, Gaussian splatting, unstructured mesh encoding, and AI-driven optimization, the proposed system will provide a new class of interactive and immersive visualization tools capable of revealing and unraveling structure, relationships, and multivariate phenomena that are inaccessible through existing rendering methods.

1. Background and Motivation

Scientific simulations and large-scale experimental facilities increasingly produce volumetric datasets of unprecedented size, resolution, and structural complexity. Many of these datasets, particularly those generated by high-performance computing environments, are defined on unstructured meshes containing irregular tetrahedral, polyhedral, or mixed-cell geometries. Conventional direct volume rendering methods, including GPU ray-marching and hierarchical empty-space skipping, have been extensively studied for these settings over the past two decades (e.g., Weiler et al., 2003; Silva et al., 2005; Morrical et al., 2023). Additionally, recent XR systems for volumetric visualization, such as medical data, demonstrate that immersive 3D environments can significantly enhance spatial understanding and user comprehension during volumetric data exploration (Liu et al., 2025), but these systems operate on pre-segmented structured volumes and do not address the rendering, data management, or interaction challenges posed by large-scale unstructured scientific datasets.

While these methods remain foundational, they increasingly struggle to deliver interactive or real-time performance for modern unstructured grids containing tens of millions of elements, especially under the additional constraints required for immersive VR systems, such as low latency, dual-eye stereo rendering, and constant framerate guarantees. Recent work in machine-learning (ML) volume representations has demonstrated remarkable gains in compression, rendering efficiency, and flexibility for structured volumetric data (Müller et al., 2022).

Early neural-field approaches such as Weiss, 2022, and Weiss & Westermann, 2022, reconstructed scalar fields using differentiable volume rendering and implicit neural networks. Subsequent systems (Lu et al., 2021; Wu et al., 2024) introduced multi-resolution hash encoding and tensor-core inference to accelerate neural volume decoding for structured grids, achieving real-time frame rates in some cases. However, these representations assume regular grid topologies, or are explicitly built around dense grids, and do not extend naturally to the

irregular connectivity or highly variable spatial density of unstructured meshes commonly used in research institutions.

In parallel, the graphics community has developed **3D Gaussian Splatting** as a high-performance representation for real-time scene reconstruction and view synthesis (Kerbl et al., 2023). Complementary explicit radiance-field approaches (Yu et al., 2021) demonstrated that high-quality view synthesis can also be achieved without neural networks by optimizing per-voxel features. Although NeRF-based radiance-field methods have driven much of this recent progress in real-time view synthesis, they are fundamentally designed for natural-scene appearance modeling rather than scientific scalar-field visualization (Xu et al., 2023).

These methods demonstrate exceptional rendering speed and quality but were designed for radiance fields, not scientific scalar fields. More importantly, existing 3DGS pipelines store view-dependent color and opacity, making them incompatible with dynamic transfer-function editing—a core requirement in scientific visualization. Very recent work by Dyken et al., 2025 introduces **Volume Encoding Gaussians** (VEG), the first Gaussian representation tailored for scientific datasets, capable of representing both structured and unstructured volumes by encoding only scalar values rather than baked-in color. VEG achieves extreme compression and high-fidelity rendering, and it supports transfer-function-agnostic visualization. However, VEG is not designed for VR, does not address stereo rendering, does not guarantee low-latency performance, and does not support interactive transformations, multivariate data, or real-time transfer-function editing. These limitations present a significant opportunity for advancing state-of-the-art scientific visualization for unstructured data.

Simultaneously, the computational imaging and graphics communities, including work from Wetzstein’s group at Stanford, have demonstrated rapid progress in Gaussian-based representations, implicit neural fields, and fast differentiable rendering, spanning topics such as large-scale Gaussian reconstruction (Xu et al., 2024), textured Gaussian scene representations (Chao et al., 2025), mixture-of-fields models for heterogeneous data, and efficient neural 3D reconstruction pipelines for microscopy and cryo-EM (Levy et al., 2023). Although these techniques point toward a broader future in learned volumetric representation, none directly address the real-time constraints, multivariate requirements, or interactive manipulation demands of VR-based unstructured scientific volume visualization.

Taken together, the field now stands at a pivotal moment. Classical GPU ray-marchers cannot meet the VR performance requirements for large unstructured datasets. Neural representations excel at compression and structured-grid reconstruction but cannot yet handle unstructured meshes or real-time VR interaction. Gaussian splatting provides exceptional rendering speed, but all existing methods, such as VEG, lack stereo VR integration, dynamic multivariate transfer-function support, and real-time spatial transformations.

These limitations illustrate a clear disconnect between the capabilities of current rendering, neural-field, and Gaussian-based approaches and the real-time, interactive demands of immersive VR exploration of unstructured scientific data. The following section outlines the specific research gaps we aim to address to bridge this divide.

2. Research Gaps

Beyond achieving real-time performance, this project directly addresses five critical gaps in current research: the absence of a VR framework for unstructured Gaussian volumes; the lack of multivariate, transfer-function-agnostic Gaussian volume exploration; the inability of existing splat-based methods to maintain stable real-time VR framerates in stereo rendering; the lack of illumination and gradient-aware shading models for unstructured Gaussian fields; and the absence of AI-driven acceleration for unstructured Gaussian rendering. Our proposed system will introduce novelties that establish a foundation for long-term research: **(1)** a unified VR engine for interactive exploration of unstructured volumes; **(2)** an optimized unstructured-to-Gaussian transcoder; **(3)** a multivariate Gaussian representation enabling multi-channel, real-time transfer-function interaction; **(4)** a low-latency stereo renderer integrated with OpenXR to ensure high-stability performance; and **(5)** an extensible AI-assisted acceleration pathway, beginning with lightweight Gaussian pruning and clustering in the first year and expanding further to neural sampling, prediction, and streaming in later phases. Together, these capabilities create a scalable pathway toward a five-year research agenda while delivering a practical, high-impact visualization system within the first year.

3. Goals

Goal 1. Real-time VR rendering of unstructured volumetric data using Gaussian splatting: This project aims to develop a novel VR-based Volume-Rendering system that supports real-time visualization of unstructured volumetric datasets through a Gaussian-splatting-based pipeline designed to maintain stable stereo frame rates in real scenes. The proposed system will integrate OVR with OpenXR to produce low-latency, high-fidelity stereoscopic views of large unstructured volumes, ensuring reliable interactivity while navigating and inspecting complex scientific data.

Goal 2. A volume-to-Gaussian transcoder for unstructured grids: We will develop a dedicated transcoding tool that transforms arbitrary unstructured grid data—such as tetrahedral meshes and adaptive, irregular geometries—into optimized Gaussian primitives. This transcoder will produce shape-adapted, spatially coherent splats that accurately encode scalar fields while remaining compact enough for real-time VR rendering. It will also support configurable density control for scalable memory and performance trade-offs. The proposed transcoder will prioritize adaptive splat shaping, scalar-field preservation, efficient mapping from unstructured vertices and cells to Gaussians, and real-time streaming to the VR renderer.

Goal 3. Interactive spatial transformations on Gaussian-encoded volumes: The proposed system provides seamless VR interaction capabilities, enabling users to manipulate Gaussian-represented volumes directly in three-dimensional space. This includes real-time rotation, translation, scaling, slicing, and region-of-interest selection, all executed at VR framerate guarantees. These interactions will operate on the splat representation itself, ensuring smooth and uninterrupted manipulation even for extremely large datasets. Achieving this requires an

optimized GPU path for manipulating millions of splats, foveated stereo-aware splat culling, spatial level-of-detail (LOD) strategies, and optional predictive adjustments based on user motion to sustain frame-rate stability.

Goal 4. Real-time, transfer-function–agnostic and multivariate visualization: We will support dynamic transfer-function editing during VR exploration, allowing users to adjust color and opacity mappings at runtime without retraining or preprocessing. The system will extend Gaussian representations to support multivariate data by attaching multiple scalar or vector attributes to each splat, enabling multi-channel transfer functions for scientific datasets containing several physical variables. This multivariate capability will allow users to uncover relationships between variables in ways not achievable with prior Gaussian or neural-field methods. The proposed system will allow scientists to modify transfer functions spontaneously while maintaining stable VR performance and consistent visual fidelity across multiple data channels.

4. Preliminary Results and Supporting Infrastructure

The VIDi Lab has established several foundational components that directly support the feasibility of the proposed VR Gaussian-volume-rendering system. The VIDi Lab’s Open Volume Renderer (OVR) provides a mature GPU-accelerated framework for volume ray-casting, supporting structured and unstructured data formats through CUDA backends and modular renderer interfaces as shown in Figure 1. OVR has been successfully used for high-fidelity rendering of data such as turbulent flows, demonstrating stability and adaptability for advanced visualization tasks. Recent extensions include a stereo-enabled OVR prototype integrated with OpenXR, enabling low-level communication between GPU render targets and VR buffering. This prototype validates OVR’s ability to drive stereoscopic rendering pipelines with interactive framerate stability, establishing the technical basis for real-time VR deployment.

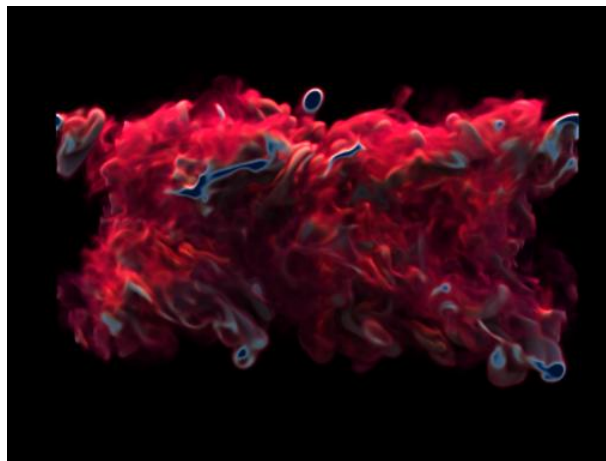


Figure 1. Preliminary rendering study using OVR: turbulent volumetric field rendered with custom transfer-function and shading pipeline. This demonstrates the renderer’s capability to handle dense volumetric structures with high visual fidelity (VIDILabs).

Additionally, techniques such as foveated importance sampling and neural-guided culling (e.g., FoVolNet; Bauer and Ma, 2023) show that VR rendering workloads can be significantly reduced

using perceptually guided sampling strategies, which directly informs our planned AI-accelerated Gaussian pruning and level-of-detail methods. Furthermore, this confirms both the feasibility and the necessity of adaptive splatting strategies for sustaining VR framerates.

These components provide a validated foundation for the proposed system. The early VR experiments confirm that interactive volume rendering in VR is technically viable within our environment, and the existing OVR/OpenXR integration ensures that our efforts in expanding to Gaussian-based unstructured data rendering can be rapidly incorporated into a full VR workflow.

5. Methodology and Proposed Tasks

The methodology proposed for this project integrates high-performance rendering, Gaussian-based data representation, VR interaction techniques, and AI-enabled acceleration. Each research task supports one or more of the four Goals and contributes to the nine Expected Deliverables:

Task 1. VR Rendering Framework for Unstructured Gaussian Volumes: We will extend OVR with a stereo-capable rendering backend that interfaces directly with OpenXR – building upon the prototype described in the Preliminary Results and Supporting Infrastructure section. The rendering pipeline will be adapted to accept Gaussian primitives as its primary data representation, enabling low-latency stereoscopic visualization. Initial tasks include integrating OpenXR session management, implementing dual-eye rendering paths, and optimizing memory transfers for VR frame stability (**Goals 1 & 3; Deliverables 1, 5**).

Task 2. Unstructured-to-Gaussian Transcoding Pipeline: We will develop a preprocessing stage that converts unstructured grids, such as tetrahedral meshes and adaptive multiresolution datasets, into optimized Gaussian primitives. This tool will compute Gaussian means, covariances, densities, and attribute sets for each cell or region, ensuring scalar-field preservation. Adaptive Gaussian placement, density control, and error-driven refinement will be explored to balance fidelity and VR rendering costs. The transcoder will be designed as a modular tool, enabling later AI-driven improvements such as neural clustering or learned splat prediction (**Goal 2; Deliverables 2, 7**).

Task 3. Real-Time VR Interaction and Transformation Module: We will construct an interaction layer that directly manipulates the Gaussian representation in VR. This includes GPU-accelerated transformations: translation, scaling, rotation, as well as slicing, clipping planes, and region-of-interest selection. The approach leverages quaternion-based transformations and instance-level Gaussian updates to preserve interactivity during user motion. Performance-critical code paths will be optimized using persistent GPU kernels and batched updates to ensure each interaction operates within VR latency constraints (**Goal 3; Deliverables 3 & 5**).

Task 4. Multivariate Transfer-Function System: A flexible transfer-function engine will be developed to operate directly on Gaussian attributes, supporting both univariate and multivariate visual mappings. Real-time transfer function updates will be implemented by re-

evaluating Gaussian opacity and color contributions without re-transcoding the dataset. For multivariate datasets, each Gaussian stores a compact vector of physical variables, enabling multi-channel transfer function design in VR. Gradient-aware shading models will be investigated to enhance structural visibility/interpretability, using local scalar derivatives computed from Gaussian covariance information (**Goal 4; Deliverables 4, 6**).

Task 5. Illumination and Gradient-Aware Shading Models: We will explore lightweight scientific shading models tailored to Gaussian volume data. Techniques include gradient-based shading, isosurface-inspired illumination, and analytic opacity modulation derived from Gaussian covariances. These methods will be tested for clarity, performance, and suitability for VR environments, ensuring visual cues remain stable across stereo rendering and transfer-function edits (**Deliverable 6**).

Task 6. Foundational AI/ML Acceleration Pathway: To enable long-term scalability, we will initiate a foundation for AI-driven optimization:

- **Gaussian pruning and reduction:** ML models will identify redundant or perceptually insignificant Gaussians, enabling aggressive simplification without degrading scientific meaning.
- **Neural clustering and importance sampling:** learned similarity metrics will group Gaussians by shared features (e.g., local gradients, scalar distributions), improving rendering efficiency.
- **Predictive LOD and streaming:** lightweight neural predictors will estimate visibility or shading importance during head motion, enabling precomputation and selective rendering.

These early models will be integrated into the transcoder and renderer as optional acceleration modes, establishing the groundwork for more advanced neural approaches in later developmental phases (**Deliverables 7 & 9; Goals 2 & 4**).

Task 7. Integration, Software Packaging, and Evaluation: The proposed system components, which include the renderer, transcoder, interaction module, transfer function engine, shading system, and AI-driven optimizations, will be integrated into a unified software prototype. Detailed benchmarking will also be conducted on representative unstructured datasets (e.g., turbulence and fluid simulations) to evaluate stability, visual fidelity, transcoder accuracy, and responsiveness under VR interaction. A final technical report will summarize performance metrics, design insights, and pathways for future developments (**Deliverables 8 & 9; All Goals**).

6. Expected Outcomes and Deliverables

Outcome 1. A functional VR engine for real-time rendering of unstructured Gaussian volumes: This outcome delivers an integrated OVR + OpenXR pipeline capable of stable stereoscopic visualization of unstructured Gaussian volumes. It directly fulfills **Goal 1**, establishing the real-time VR rendering framework required for scientific data exploration. It addresses **Gap 1** (no VR framework for unstructured Gaussian volumes) and **Gap 3** (lack of stable stereo VR rendering).

Outcome 2. A deployable unstructured-to-Gaussian transcoder for large-scale datasets

We will provide an innovative transcoding tool that converts unstructured simulation or experimental fields, tetrahedral meshes, irregular grids, or AMR structures, into optimized Gaussian primitives suitable for VR. This satisfies **Goal 2** and addresses **Gap 1** (representation needed for VR), **Gap 2** (transfer-function-agnostic multivariate support via scalar mapping), and **Gap 5** (enabling future AI acceleration). The transcoder aims towards the inclusion of density controls, adaptive splat shaping, and support for real-time streaming.

Outcome 3. A complete VR interaction module for transforming Gaussian-encoded volumes in 3D space:

This includes real-time interaction capabilities such as rotation, translation, scaling, slicing, and region-of-interest selection performed directly on the Gaussian representation. It directly fulfills **Goal 3** and addresses **Gap 3** by ensuring all transformations remain within VR frame stability constraints. By enabling direct manipulation of splats, this also partially reduces **Gap 1**, completing the VR workflow for unstructured Gaussian data.

Outcome 4. A real-time, multivariate transfer-function system integrated into VR: This system enables users to edit transfer functions on the fly and explore multivariate data by attaching multiple scalar or vector attributes to each Gaussian. It fulfills **Goal 4** and directly addresses **Gap 2**. The editor will support smooth updates without re-rendering stalls, enabling scientific hypothesis-driven exploration of complex variables.

Outcome 5. A low-latency stereo Gaussian renderer optimized for VR headsets: This deliverable includes the VR-optimized splat renderer with performance features such as foveated culling, adaptive LOD, and predictive motion adjustments. It satisfies **Goals 1 and 3** and directly addresses **Gap 3** by enabling real-time VR framerate optimization in stereo rendering under transformations, transfer function edits, and navigation. This renderer forms the computational core of the system.

Outcome 6. A prototype illumination and gradient-aware shading module for Gaussian volumes: We develop gradient-based shading to improve interpretability for scientific fields by exploring shading and illumination methods. This addresses **Gap 4** (lack of illumination models for Gaussian volumes) and enhances the visual fidelity of all goals, especially **Goal 1** (rendering) and **Goal 4** (multivariate visualization).

Outcome 7. A foundational AI-enabled acceleration pipeline: This includes ML integrations such as Gaussian pruning, clustering for variable reduction, and preliminary neural importance sampling. It directly addresses **Gap 5** (absence of AI acceleration) and creates a scalable pathway toward advanced neural acceleration in years 2–5. It supports **Goal 2** (future transcoder improvements) and **Goal 4** (future multivariate optimization).

Outcome 8. A unified software package and documentation: We will deliver a packaged prototype containing the VR renderer, transcoder, interaction system, shading models, and baseline AI pipeline, along with developer documentation and instructions for integration with our visualization workflow. This satisfies **all four Goals** by providing a functional, end-to-end system and establishes the foundation for a multi-year development cycle.

Outcome 9. Evaluation results and a technical report summarizing system performance: This includes benchmarking VR frame stability, renderer throughput, Gaussian-encoding quality, multivariate interaction responsiveness, and shading interpretability. It verifies that we have met **Goal 1** (real-time VR), **Goal 2** (transcoder quality), **Goal 3** (interactive transformation stability), and **Goal 4** (multivariate transfer function editing). It also provides quantitative evidence addressing all **Gaps 1–5**.

7. Researcher’s Preparation and Qualifications

Michael Martin has a strong interdisciplinary background at the intersection of virtual reality, real-time rendering, AI/ML, and scientific visualization, with multiple research projects that directly prepare him for this proposed work (Martin, 2025). At UC Davis’s VIDi Lab, Martin worked on **ProneVR**: an immersive therapeutic VR system that integrates real-time rendering, shader-based effects, interactive simulation, and 3D spatial acoustics (Martin et al., 2025). Previously, he developed a VR-based ophthalmology prototype for visual field testing, demonstrating his ability to design medically oriented immersive systems requiring precision, real-time responsiveness, and scientific validity. His experience further extends to GPU-accelerated 4D volume rendering using CUDA-based pipelines, providing hands-on expertise in performance-critical rendering paths relevant to the real-time requirements of this proposal. Beyond VR, his research spans ML, deep learning, explainable AI – including work on diffusion models – and scientific visualization on complex large-scale datasets funded by NASA at the Desert Research Institute, establishing a strong foundation for the AI-enabled components. Martin also designed and developed interactive systems, spatial simulation pipelines, and real-time environments in Unity and custom rendering frameworks, demonstrating the technical breadth necessary to execute and extend this proposed VR Gaussian-volume rendering research.

8. Broader Impacts

The proposed project has the potential to advance scientific discovery, broaden participation in computational science, and improve access to complex scientific data across disciplines. Modern research fields such as fluid dynamics, climate modeling, astrophysics, neuroscience, combustion, and materials science increasingly generate unstructured volumetric datasets that are too large and too complex for traditional 2D or static 3D visualization tools. By enabling real-time, immersive exploration of these datasets, this project enables the identification of patterns, structural formations, and multivariate relationships that are otherwise difficult or impossible to discern. This can accelerate hypothesis generation, improve interpretability of simulation results, and support more rapid iteration in scientific workflows.

The project also contributes to the broader scientific computing ecosystem by developing methods that integrate AI acceleration with interactive visualization. The techniques developed (AI-guided sampling, adaptive Gaussian splatting, and real-time VR rendering) can serve as a foundation for next-generation visualization systems at national laboratories, supercomputing

centers, and university research facilities. These innovations may further support real-time analysis of streaming simulation outputs, high-dimensional multivariate scientific data, and data-intensive experiments such as light-source, plasma physics, or turbulent flow measurements. Over time, the approach offers a pathway toward immersive steering of simulations and model-in-the-loop optimization, opening new modes of scientific engagement.

This project will also enhance education and workforce development by providing a modern platform for training students in visualization, VR development, AI-driven rendering, and unstructured data analysis, which are skills that are increasingly essential and high in demand across computational science, engineering, and data-driven research fields. Students will gain hands-on experience with advanced state-of-the-art GPU programming, XR interfaces, scientific visualization pipelines, and AI models for performance optimization. The project will produce open-source software and documentation that can be used in research labs and classrooms.

Additionally, Immersive visualization also offers accessibility improvements for learners who benefit from spatial, interactive, or 3D learning modalities. By transforming abstract numerical fields into intuitive spatial experiences, this proposed project can support educational outreach, virtual field trips, and public STEM demonstrations that increase scientific literacy and broaden participation in computing-related disciplines.

Finally, the project strengthens cross-disciplinary collaboration. The resulting framework will enable domain scientists, computer scientists, and data analysts to explore unstructured datasets collaboratively within a shared VR space. This supports the ultimate goals of fostering interdisciplinary research, advancing data-intensive science, and democratizing access to emerging AI-accelerated immersive visualization technologies.

References

- Bauer, D., Wu Q, Ma KL. FoVolNet: Fast Volume Rendering using Foveated Deep Neural Networks. *IEEE Trans Vis Comput Graph*. 2023 Jan;29(1):515-525. doi: 10.1109/TVCG.2022.3209498. Epub 2022 Dec 19. PMID: 36155446; PMCID: PMC10984251.
- Chao, B., Tseng, H.-Y., Porzi, L., Gao, C., Li, T., Li, Q., Saraf, A., Huang, J.-B., Kopf, J., Wetzstein, G., & Kim, C. (2025). Textured Gaussians for Enhanced 3D Scene Appearance Modeling. arXiv:2411.18625. <https://arxiv.org/abs/2411.18625>
- Dyken, C., Gajjar, S., Chan, M., & Owada, S. (2025). Volume Encoding Gaussians: Transfer-Function-Agnostic 3D Gaussians for Volume Rendering. arXiv:2504.13339. <https://arxiv.org/html/2504.13339v1>
- Kerbl, B., Kopanas, G., Leimkühler, T., & Drettakis, G. (2023). 3D Gaussian Splatting for Real-Time Radiance Field Rendering. arXiv:2308.04079. <https://arxiv.org/abs/2308.04079>
- Levy, A., Raghu, R., Shustin, D., Peng, A. R.-Y., Li, H., Clarke, O. B., Wetzstein, G., & Zhong, E. D. (2024). Mixture of Neural Fields for Heterogeneous Reconstruction in Cryo-EM. arXiv:2412.09420. <https://arxiv.org/abs/2412.09420>

- Liu, Q., Qiu, S., Wang, Y., Wu, X., Chok, K. S. H., Fu, C.-W., & Heng, P.-A. (2025). Coordinated 2D–3D Visualization of Volumetric Medical Data in XR with Multimodal Interactions. arXiv:2506.22926. <https://arxiv.org/abs/2506.22926>
- Lu, Y., Jiang, K., Levine, J. A., & Berger, M. (2021). Compressive Neural Representations of Volumetric Scalar Fields. arXiv:2104.04523. <https://arxiv.org/abs/2104.04523>
- Martin, M.R. (2025). Personal Website. <https://michaelmartintech.github.io/>
- Martin, M., Chan, G., Ma, K.-L. (2025). ProneVR: Underwater Demo [Computer software]. GitHub. <https://github.com/MichaelMartinTech/ProneVR-Demo>
- Morriscal, N., A. Sahistan, U. Güdükbay, I. Wald and V. Pascucci, "Quick Clusters: A GPU-Parallel Partitioning for Efficient Path Tracing of Unstructured Volumetric Grids," in IEEE Transactions on Visualization and Computer Graphics, vol. 29, no. 1, pp. 537-547, Jan. 2023, doi: 10.1109/TVCG.2022.3209418.
- Müller, T., Evans, A., Schied, C., & Keller, A. (2022). Instant Neural Graphics Primitives with a Multiresolution Hash Encoding. ACM Transactions on Graphics, 41(4), 1–15. <https://doi.org/10.1145/3528223.3530127>
- Silva, Claudio & Comba, João & Callahan, Steven & Bernardon, Fábio. (2005). A Survey of GPU-Based Volume Rendering of Unstructured Grids. RITA. 12. 9-30.
- VIDILabs (Bauer, D., Wu, Q., Ma, K.-L.). (2023). Open Volume Renderer (OVR) [Computer software]. GitHub. <https://github.com/VIDILabs/open-volume-renderer>
- Weiler, M., M. Kraus, M. Merz, and T. Ertl, "Hardware-based ray casting for tetrahedral meshes," IEEE Visualization, 2003. VIS 2003. Seattle, WA, USA, 2003, pp. 333-340, doi: 10.1109/VISUAL.2003.1250390.
- Weiss, S., & Westermann, R. (2022). Differentiable Direct Volume Rendering. IEEE Transactions on Visualization and Computer Graphics, 28(1), 562–572. <http://dx.doi.org/10.1109/TVCG.2021.3114769>
- Weiss, Sebastian & Hermüller, P. & Westermann, R.. (2022). Fast Neural Representations for Direct Volume Rendering. Computer Graphics Forum. 41. 10.1111/cgf.14578.
- Wu, Q., D. Bauer, M. J. Doyle and K. -L. Ma, "Interactive Volume Visualization via Multi-Resolution Hash Encoding Based Neural Representation," in IEEE Transactions on Visualization and Computer Graphics, vol. 30, no. 8, pp. 5404-5418, Aug. 2024, doi: 10.1109/TVCG.2023.3293121.
- Xu, L., Vasu Agrawal, William Laney, Tony Garcia, Aayush Bansal, Changil Kim, Samuel Rota Bulò, Lorenzo Porzi, Peter Kotschieder, Aljaž Božič, Dahua Lin, Michael Zollhöfer, and Christian Richardt. 2023. VR-NeRF: High-Fidelity Virtualized Walkable Spaces. In SIGGRAPH Asia 2023 Conference Papers (SA '23). Association for Computing Machinery, New York, NY, USA, Article 43, 1–12. <https://doi.org/10.1145/3610548.3618139>

Xu, Y., Shi, Z., Yifan, W., Chen, H., Yang, C., Peng, S., Shen, Y., & Wetzstein, G. (2024). GRM: Large Gaussian Reconstruction Model for Efficient 3D Reconstruction and Generation. arXiv:2403.14621. <https://arxiv.org/abs/2403.14621>

Yu, A., Fridovich-Keil, S., Tancik, M., Chen, Q., Recht, B., & Kanazawa, A. (2021). Plenoxels: Radiance Fields without Neural Networks. arXiv:2112.05131. <https://arxiv.org/abs/2112.05131>